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Valve: Research

Patient-Prosthesis Mismatch After Surgical Aortic Valve Replacement: Analysis of the PARTNER Trials

Vinod H. Thourani, MD,[1,*] Amr E. Abbas, MD,[2] Julien Ternacle, MD, PhD,[3,4] Rebecca T. Hahn, MD,[5,6] Raj Makkar, MD,[7] Susheel K. Kodali, MD,[5] Isaac George, MD,[8] Samir Kapadia, MD,[9] Lars G. Svensson, MD, PhD,[10] Wilson Y. Szeto, MD,[11] Howard C. Herrmann, MD,[12] Gorav Ailawadi, MD,[13] Jonathon Leipsic, MD,[14] Philipp Blanke, MD,[14] John Webb, MD,[14] Wael A. Jaber, MD,[9] Mark Russo, MD,[15] S. Chris Malaisrie, MD,[16] Pradeep Yadav, MD,[17] Marie-Annick Clavel, MD,[4] Omar K. Khalique, MD,[6,18] Neil J. Weissman, MD,[19] Pamela Douglas, MD,[20] Jeroen Bax, MD,[21] Abdellaziz Dahou, MD,[6] Ke Xu, MS,[22] Vinayak Bapat, MD,[23] Maria C. Alu, MS,[5,6] Martin B. Leon, MD,[5,6] Michael J. Mack, MD,[24] and Philippe Pibarot, DVM, PhD[4]

ABSTRACT

BACKGROUND Our objective was to compare the impact of patient-prosthesis mismatch (PPM) for 2 years after surgical aortic valve replacement within the prospective, randomized Placement of Aortic Transcatheter Valves (PARTNER) trials.

METHODS Surgical aortic valve replacement patients from the PARTNER 1, 2, and 3 trials were included. PPM was classified as moderate (indexed effective orifice area ≤0.85 cm^2/m^2) or severe (indexed effective orifice area ≤0.65 cm^2/m^2). The primary endpoint was the composite of all-cause death and heart failure rehospitalization at 2 years.

RESULTS By the predicted PPM method (PPM_P), 59.1% had no PPM, 38.8% moderate PPM, and 2.1% severe PPM; whereas by the measured PPM method (PPM_M), 42.4% had no PPM, 36.0% moderate, and 21.6% severe. Patients with no PPM_P (23.6%) had a lower rate of the primary endpoint compared with patients with moderate (28.2%, $P = .03$) or severe PPM_P (38.8%, $P = .02$). Using the PPM_M method, there was no difference between the no (17.7%) and moderate PPM_M groups (21.1%) in the primary outcome ($P = .16$). However, those with no PPM_M or moderate PPM_M were improved compared with severe PPM_M (27.4%, $P < .001$ and $P = .02$, respectively).

CONCLUSIONS Severe PPM analyzed by PPM_P was only 2.1% for surgical aortic valve replacement patients. The PPM_M method overestimated the incidence of severe PPM relative to PPM_P, but was also associated with worse outcome. There was higher all-cause mortality in patients with severe PPM, thus surgical techniques to minimize PPM remain critical.

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Patient-prosthesis mismatch (PPM) after surgical aortic valve replacement (SAVR) remains an area of debate and controversy since its introduction

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[1]Department of Cardiovascular Surgery, Marcus Valve Center, Piedmont Heart Institute, Atlanta, Georgia; [2]Department of Cardiology, Beaumont Hospital, Detroit, Michigan; [3]Department of Cardiology, Haut-Leveque Cardiology Hospital, Bordeaux University, Pessac, France; [4]Department of Cardiology, Quebec Heart & Lung Institute, Laval University, Quebec, Canada; [5]Division of Cardiology, Columbia University Irving Medical Center/New York-Presbyterian Hospital, New York, New York; [6]Cardiovascular Research Foundation, New York, New York; [7]Department of Cardiology, Cedars Sinai Medical Center, Los Angeles, California; [8]Division of Cardiac Surgery, Columbia University Irving Medical Center/New York-Presbyterian Hospital, New York, New York; [9]Department of Cardiology, Cleveland Clinic, Cleveland, Ohio; [10]Department of Cardiac Surgery, Cleveland Clinic, Cleveland, Ohio; [11]Division of Cardiac Surgery, University of Pennsylvania, Philadelphia, Pennsylvania; [12]Division of Cardiology, University of Pennsylvania, Philadelphia, Pennsylvania; [13]University of Michigan, Ann Arbor, Michigan; [14]Department of Cardiology, University of British Columbia, Vancouver, Canada; [15]Division of Cardiac Surgery, Rutgers-Robert Wood Johnson Medical School, New Brunswick, New Jersey; [16]Division of Cardiac Surgery, Northwestern University, Chicago, Illinois; [17]Department of Cardiology, Marcus Valve Center, Piedmont Heart Institute, Atlanta, Georgia; [18]Department of Cardiology, St. Francis Hospital, Roslyn, New York; [19]MedStar Health Research Institute, Washington, DC; [20]Division of Cardiology, Duke University Medical Center, Durham, North Carolina; [21]Department of Cardiology, Leiden University Medical Center, Leiden, the Netherlands; [22]Edwards Lifesciences, Irvine, California; [23]Department of Cardiac Surgery, Minneapolis Heart Institute, Minneapolis, Minnesota; and [24]Department of Cardiac Surgery, Baylor Scott and White Health, Dallas, Texas

Address correspondence to Dr Thourani, Department of Cardiovascular Surgery, Marcus Valve Center, Piedmont Heart Institute, 95 Collier Rd, Ste 5015, Atlanta, GA 30309; email: vinod.thourani@piedmont.org.

Abbreviations and Acronyms

AE = annular enlargement
AV = aortic valve
BMI = body mass index
EOA = effective orifice area
EOAi = indexed EOA
HF = heart failure
LV = left ventricular
NYHA = New York Heart Association
PARTNER = Placement of Aortic Transcatheter Valves
PPM = patient-prosthesis mismatch
PPM_M = measured PPM method
PPM_P = predicted PPM method
SAVR = surgical aortic valve replacement
TAVR = transcatheter aortic valve replacement
STS = The Society of Thoracic Surgeons
TTE = transthoracic echocardiography

by Rahimtoola[1] in 1978 and popularization by Pibarot and colleagues[2] in 2000. Commonly defined by an effective orifice area (EOA) too small in relation to the patient's body surface area, PPM is generally categorized using the indexed EOA (EOAi: the EOA divided by the patient's body surface area). PPM is defined as moderate if $\leq$0.85 cm^2/m^2 (by some definitions $\leq$0.70 cm^2/m^2 if the body mass index [BMI] is $\geq$30 kg/m^2) and severe if $\leq$0.65 cm^2/m^2 (or $\leq$0.55 cm^2/m^2 if the BMI is $\geq$30 kg/m^2).[3] While most SAVR studies have used the EOAi predicted by the manufacturer-provided reference values for a given valve type and size, transcatheter aortic valve replacement (TAVR) trials have generally used EOAi measured by 30-day postoperative transthoracic echocardiography (TTE) to determine the incidence of PPM.[4-6]

Prior studies of PPM in patients undergoing SAVR were performed retrospectively and without an echo core laboratory—and the vast majority of these studies only utilized the predicted PPM (PPM_P) method, not the measured PPM (PPM_M).[5,6] The objective of this post-hoc study was to analyze the incidence and clinical impact of PPM_P vs PPM_M at 2 years post SAVR within the prospective, randomized Placement of Aortic Transcatheter Valves (PARTNER) trials.

PATIENTS AND METHODS

STUDY POPULATION. The analysis population included severe, symptomatic aortic stenosis patients randomized to SAVR in the PARTNER trials: P1A, P2A, and P3. All 3 trials were approved by the institutional review boards of each participating site, and all patients provided written informed consent.

PPM ASSESSMENT METHODS AND DEFINITIONS. Two methods were used to assess the presence of PPM: PPM_M and PPM_P. For PPM_M, EOA was calculated at 30 days using TTE and the continuity equation.[7] Echocardiograms were analyzed independently by a consortium of echocardiography core laboratories.

For PPM_P, EOA was based on the normal reference value for each size and model of implanted surgical valve, as obtained from published data. For both methods, EOA was indexed (EOAi) to body surface area calculated using the Dubois formula. PPM was defined as insignificant if EOAi was >0.85 cm^2/m^2, moderate if >0.65 cm^2/m^2 and $\leq$0.85 cm^2/m^2, and severe if $\leq$0.65 cm^2/m^2. A sensitivity analysis was conducted using the more contemporary Valve Academic Research Consortium 3 definitions of PPM, which use adjusted thresholds for patients with BMI $\geq$30 kg/m^2: In obese patients, PPM is considered insignificant if EOAi >0.70, moderate if >0.55 cm^2/m^2 and $\leq$0.70 cm^2/m^2, and severe if <0.55 cm^2/m^2.[3]

STUDY ENDPOINTS. The primary endpoint was a composite of all-cause death and heart failure (HF) rehospitalization at 2 years. Secondary endpoints included all-cause death, cardiac death, and HF rehospitalization at 2 years. Patients were considered to be in a low flow state when stroke volume index was <35 mL/m^2.

STATISTICAL ANALYSIS. PPM incidence and association with outcomes were analyzed separately for PPM_M and PPM_P. Continuous data are presented as mean $\pm$ SD. Categorical data are presented as percentage and fraction of occurrence. Group comparisons were analyzed with the χ^2 test for categorical variables; analysis of variance was used for continuous variables. Paired comparisons were performed using McNemar's test when appropriate. Kaplan-Meier estimates and the log-rank test were used to compare occurrence of endpoints over 2 years, stratified by PPM (none, moderate, or severe), for both PPM_M and PPM_P. Multivariable analysis was performed with Cox proportional hazards model including The Society of Thoracic Surgeons (STS) score, age, sex, diabetes, and prior coronary artery bypass grafting. All statistical analyses were performed using SAS software, version 9.4 (SAS Institute), and $P < .05$ was considered statistically significant.

RESULTS

BASELINE CHARACTERISTICS. The study flowchart is presented in Figure 1 and baseline characteristics of the PPM_P and PPM_M cohorts are presented in Table 1 and Supplemental Table 1, respectively. Of the 1826 patients undergoing SAVR, PPM by predicted EOAi method was evaluable in 1682 (92.1%) patients and by measured EOAi method in 1348 (73.8%). A total of 1330 patients have both measured and predicted PPM. Among patients with evaluable PPM_P, those with severe PPM_P were younger, less likely male, more

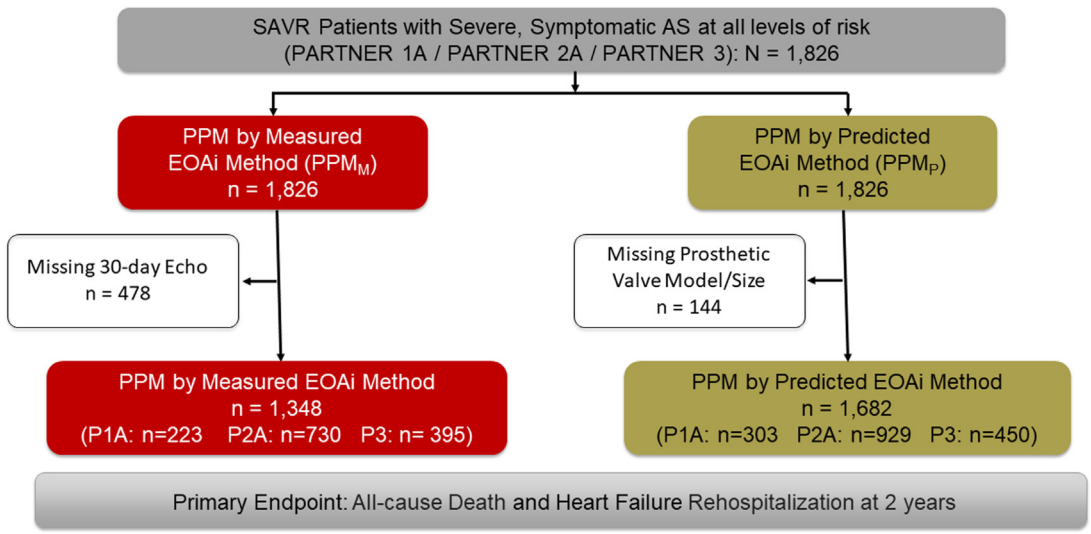

FIGURE 1 Study population according to patient-prosthesis mismatch (PPM) assessment method. (AS, aortic stenosis; Echo, echocardiogram; EOAi, effective orifice area divided by the patient's body surface area; P, Placement of Aortic Transcatheter Valves (PARTNER) trial; PPM$_M$, measured patient-prosthesis mismatch; PPM$_P$, predicted patient-prosthesis mismatch; SAVR, surgical aortic valve replacement.)

frequently in New York Heart Association (NYHA) class III/IV, had larger BMI, were more likely diabetic, and had a lower stroke volume index (Table 1). Among patients with evaluable PPM$_M$, those with severe PPM$_M$ were less likely male, had higher NYHA class III/IV, larger BMI, and more coronary artery disease, and had a lower stroke volume index (Supplemental Table 1). Of note, the prevalence of low-flow state was 2-fold higher in patients with severe PPM$_M$ (55%) vs those with no PPM$_M$ (26%).

PROCEDURAL CHARACTERISTICS. Supplemental Table 2 shows the procedural characteristics of the PPM$_P$ and PPM$_M$ groups. As expected, progressively smaller prosthetic valve sizes were associated with increasing severity of PPM$_P$: Nearly 75% of severe PPM$_P$ patients had a 19- or 21-mm valve (vs 23% of no PPM$_P$ and 61% of moderate PPM$_P$ patients). Differences in the proportion of small surgical valve implants were lower when using PPM$_M$: 57% of severe PPM$_M$ and 43% of moderate PPM$_M$ patients. The most common valve

TABLE 1 Baseline Characteristics Predicted Patient-Prosthesis Mismatch

	No PPM$_P$ (n = 994)	Moderate PPM$_P$ (n = 653)	Severe PPM$_P$ (n = 35)	Total (N = 1682)	*P* Value
Age, y	80.5 ± 7.3	79.2 ± 8.0	77.7 ± 8.6	80.0 ± 7.6	<.001
Male	65.8	49.9	60.0	59.5	<.001
NYHA class III or IV	60.3	72.4	74.3	65.3	<.001
STS-PROM score, %	5.6 ± 3.7	6.1 ± 3.9	5.9 ± 2.9	5.8 ± 3.8	.03
Body mass index, kg/m^2	27.3 ± 5.0	30.4 ± 6.4	33.3 ± 9.0	28.6 ± 6.0	<.001
Diabetes	31.6	38.7	42.9	34.6	.007
Hypertension	90.6	94.5	94.3	92.2	.02
Atrial fibrillation	29.3	30.5	27.6	29.7	.87
Coronary artery disease	60.9	62.8	57.1	61.5	.64
AV mean gradient, mm Hg	45.8 ± 12.9	44.9 ± 12.5	47.2 ± 12.7	45.5 ± 12.8	.28
LVEF, %	57.2 ± 12.5	57.2 ± 12.4	55.6 ± 12.3	57.2 ± 12.4	.75
Stroke volume index, mL/m^2	39.3 ± 9.7	37.3 ± 11.0	35.5 ± 8.6	38.4 ± 10.3	.0003
Low flow (<35 mL/m^2)	33.0	44.4	53.1	37.8	<.001
30-d low flow (<35 mL/m^2)	46.2	60.2	73.9	52.1	<.001

Values are presented as % or mean ± SD. AV, aortic valve; LVEF, left ventricular ejection fraction; NYHA, New York Heart Association; PPM$_P$, predicted patient-prosthesis mismatch; PROM, predicted risk of mortality; STS, The Society of Thoracic Surgeons.

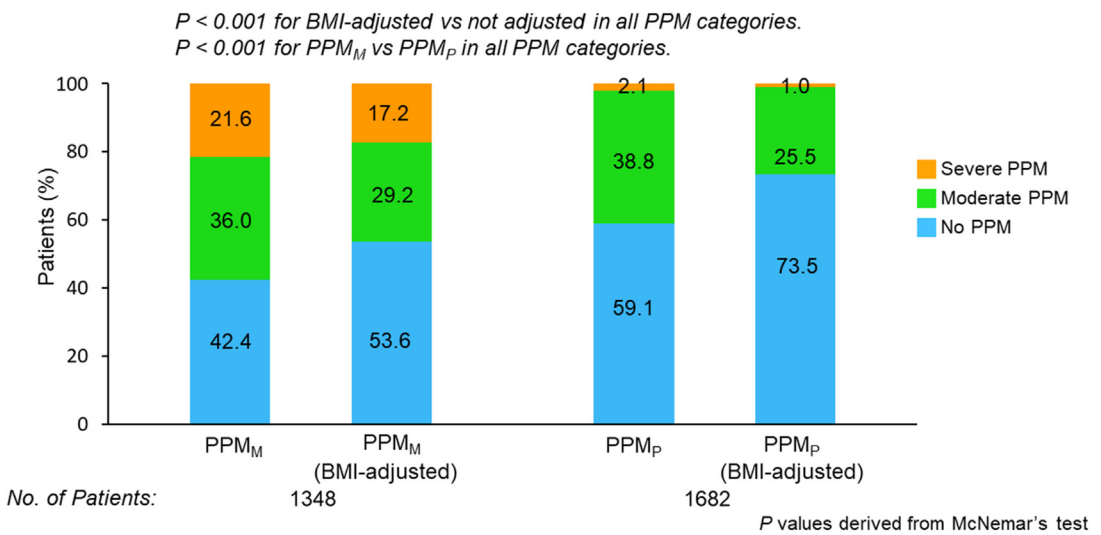

FIGURE 2 Incidence of measured patient-prosthesis mismatch (PPM_M) and predicted patient-prosthesis mismatch (PPM_P) (with and without adjusting for body mass index [BMI]).

implanted in both the PPM_P and PPM_M groups was an internally mounted pericardial bioprosthesis (~80% in each group) (Supplemental Table 3). Patients receiving the stentless/sutureless bioprosthesis had the lowest percent of severe PPM, regardless of method.

The distribution of PPM by each method, with and without obesity-adjusted thresholds, is shown in Figure 2. When patients with severe PPM_M were reclassified using the PPM_P method, 5.2% remained as severe PPM, while 64.8% were deemed to be moderate and 30% to have no PPM (Supplemental Figure 1).

ECHOCARDIOGRAPHIC FINDINGS. As expected, mean aortic valve (AV) gradients were highest in the severe PPM_P group at 30-day echocardiography; this was maintained through 2 years. The gradient in this group was on average ~7 mm Hg higher vs the no PPM_P group and ~4 mm Hg vs the moderate PPM_P group (Figure 3A). There were minimal increases in mean gradients from 30 days to 2 years (~1 mm Hg) in all groups. Similarly, mean AV gradients were highest in the severe PPM_M group from 30 days until 2 years, but these differences were less pronounced than by the PPM_P method: ~5 mm Hg severe PPM_M vs no PPM_M and ~1 mm Hg vs moderate PPM_M (Figure 3B). At 2 years, there was no significant difference in left ventricular (LV) mass index between PPM groups in either the PPM_P or PPM_M cohort (Supplemental Tables 4, 5).

SURVIVAL. Figure 4 shows Kaplan-Meier curves for the primary endpoint. In patients analyzed with the PPM_P method, those with no PPM had significantly lower rates of mortality or HF rehospitalization (23.6%) compared with those with moderate (28.2%, $P = .03$) or severe PPM_P (38.8%, $P = .02$) (Figure 4A). Although there was a 10% absolute difference in the primary outcome between those with moderate vs severe PPM, this was not statistically different ($P = .14$). Using the PPM_M method, there was no difference between the no PPM_M (17.7%) and moderate PPM_M (21.1%) groups in the primary outcome ($P = .16$) (Figure 4B). However, those with no PPM_M or moderate PPM_M were significantly improved compared with the severe PPM_M group (27.4%, $P < .001$ and $P = .02$, respectively). Individual Kaplan-Meier curves for all-cause and cardiovascular mortality and HF rehospitalization are presented in Supplemental Figures 2-7. At 2 years, 1 patient underwent repeat SAVR in the severe PPM_P group and 1 underwent repeat SAVR and 1 underwent TAVR-in-SAVR in the severe PPM_M group.

Multivariable analysis for the primary endpoint and individual components is noted in Supplemental Table 6 for PPM_P and PPM_M cohorts. Severe PPM_P was an independent predictor of the primary composite and all-cause death. Higher STS score was predictive of all outcomes at 2 years. Severe PPM_M was an independent predictor of the primary composite outcome, all-cause death, and HF rehospitalization.

COMMENT

In these SAVR patients, we evaluated the prevalence and outcomes of none, moderate, or severe PPM at 2 years utilizing predicted and measured EOAi. The main findings are as follows: First, severe PPM analyzed by

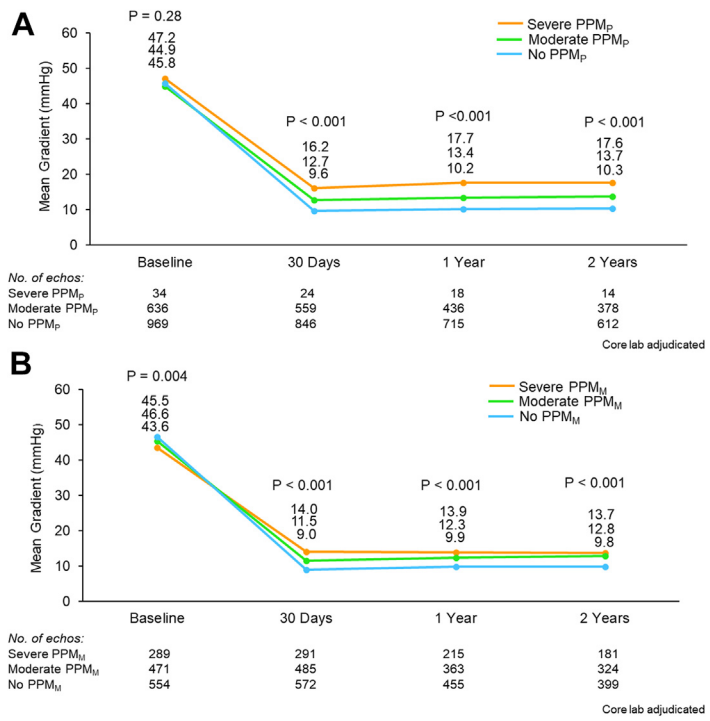

FIGURE 3 (A) Mean aortic valve gradient, predicted patient-prosthesis mismatch (PPM_P). (B) Mean aortic valve gradient, measured patient-prosthesis mismatch (PPM_M). (echos, echocardiograms.)

the predicted EOAi was much (10-fold) lower when compared with the measured EOAi (2.1% vs 21.6%, respectively). Second, BMI adjustment of EOAi cutpoints in obese individuals revealed further reduction in severe PPM for predicted and measured EOAi (1.0% vs 17.2%, respectively). Third, the implantation of small surgical valves (19 or 21 mm) was overwhelmingly associated with severe PPM (PPM_P group:75.0% and PPM_M group: 57.0%). Fourth, the presence of severe PPM was apparent at the 30-day echocardiogram and maintained through 2 years, with only minimal increase in mean gradients from 30 days to 2 years (~1 mm Hg). Fifth, using either predicted or measured EOAi, patients with no PPM had significantly improved survival and reduced HF rehospitalizations at 2 years compared with those with severe PPM. Lastly, the PPM_P method is more reflective of the impact of severe PPM on hemodynamic (AV gradients) and clinical outcomes and more specific for true-severe PPM, while the PPM_M method is less able to differentiate clinically and hemodynamically significant PPM, as evidenced by similar mean gradients between moderate and severe PPM_M; nonetheless, it, too, was associated with adverse 2-year outcome, though this may be related, in part to a low-flow state.

The present study, like previous studies,[4] suggests that measured EOAi may grossly inflate the reported incidence of severe PPM because of pseudo-severe PPM related to low-flow state. In this cohort of patients, nearly 40% were in low-flow state. In low-flow state, the leaflets of both native aortic valves and bioprosthetic aortic valves may not open completely and may therefore result in pseudo-severe PPM when EOA is measured by TTE. This pseudo-severe PPM phenomenon may explain why the difference in gradients between severe PPM_M group and moderate or no PPM_M groups is small and probably clinically insignificant. Furthermore, the impact of severe PPM_M on clinical outcomes is likely related partially to true-severe PPM, but also partially to underlying low-flow state, which has been associated with worse outcome both pre and post TAVR. Additionally, in a previous analysis compared to normal flow state, severe PPM_M was more common after both SAVR and TAVR in patients with low-flow state and was associated with worse outcomes.[4]

The predicted EOAi method overcomes this issue of pseudo-severe PPM_M and thus provides a more accurate assessment of the incidence of true-severe PPM and its specific impact on outcomes. However, to obtain an accurate predicted EOAi, it is essential to know the exact model and label size of the bioprosthetic valve and to use reliable sources for the normal reference values of EOA for SAVR valves. The predicted EOAi appears to be the preferred methodology to identify PPM because it is more specific to the PPM per se and independent of a patient's flow status.

PRIOR SAVR STUDIES. With the increase in implantation of bioprosthetic valves to ~90% of SAVR in the United States,[8] the importance of PPM as first described 44 years ago by Rahimtoola[1] remains of interest to surgeons and cardiologists. While some studies have shown increased mortality, worse NYHA functional class, and less regression of LV mass with severe PPM,[5] others have reported similar mortality and regression of LV hypertrophy. These differences may be due to discrepant definitions of PPM (ie, indexed geometric orifice area vs EOAi); the method of estimating EOAi (predicted or measured); lack of an independent, echocardiographic core laboratory if using the measured method; types and sizes of prosthesis implanted; and heterogenous populations.[9] Fallon and associates[6] reported that among 59,779 patients in the STS database from 2004 to 2014, moderate PPM occurred in 54% and severe PPM in 11%. They noted that moderate and severe PPM were associated with significantly increased risk of re-admission for heart failure (HR 1.15 and HR 1.37, respectively) and redo-SAVR (HR 1.41 and HR 2.68, respectively). They noted that survival was significantly worse for any degree of PPM with 10-year adjusted survival rates of 46%, 43%, and 35% for none, moderate, and severe PPM, respectively ($P < .001$). In a

meta-analysis of 58 articles and 40,381 patients, Dayan and colleagues[9] have noted that moderate and severe PPM were associated with a 1.5- and 2.5-fold increase in 30-day mortality after SAVR. However, only severe PPM was associated with 1.4-fold increased risk in overall mortality. Predictors of PPM were old age, female sex, hypertension, diabetes, renal failure, larger body surface area, larger BMI, and the utilization of a bioprosthesis.[9] Moreover, they and others have noted that the impact of PPM on early mortality was more prominent in those patients ≤70 years.[10]

The increased risk of late mortality associated with severe PPM may be due to the persistence of LV hypertrophy, impaired coronary flow reserve, and/or the persistence/recurrence of heart failure due to the residual LV afterload related to PPM[10]; the persistence of concomitant untreated other valve disease[9]; and the accelerated structural degeneration of bioprosthetic valves.[5,9-11]

PREVENTION OF PPM. Severe PPM after SAVR may be avoided by preventive strategies during surgery, and surgeons should remain hypervigilant to prevent this avoidable situation. In cases where severe PPM cannot be avoided, aortic root or annular enlargement (AE) is the most accepted technique for preventing severe PPM. A meta-analysis comparing 2570 SAVR plus concomitant aortic root enlargement patients with 5991 SAVR-only patients showed no difference in early mortality despite an increase of 20 minutes cardiopulmonary bypass time and 14 minutes aortic crossclamp time.[12] Although high-volume surgeons do not hesitate to perform AE, it remains an uncommonly performed procedure by most surgeons. Hawkins and coworkers[13] reported that only 5% patients underwent AE in their series, which varied immensely by individual hospital. However, AE was associated with increased operative mortality (4.7% vs 2.5%, $P = .24$; odds ratio, 2.06, $P = .016$) and major morbidity (odds ratio, 1.41, $P = .42$). They concluded that AE should be weighed against the increased perioperative risk. Mehaffey and colleagues[14] reviewed 5412 of 189,268 (2.9%) patients undergoing AE from 2008 to 2016 in the STS database. They noted that patients undergoing AE had higher risk-adjusted rates of 30-day complications and death (5.4% vs 3.4%, $P < .0001$). They showed a higher risk of mortality with AE during the first 3 years, after which the curves crossed, significantly favoring AE. These studies highlight the critical need for standardization, education, and quality improvement for AE techniques due to high variability in surgeon expertise and increased early mortality.

Moreover, surgeons with less experience in performing an AE may wish to obtain a preoperative computed tomography prior to SAVR.[15] If it is anticipated that the patient will require AE or other more sophisticated surgical options; the surgeon will be prepared in advance or could refer to a surgeon with aortic root reconstruction expertise. Surgeons may also want to consider a mechanical valve choice as an additional strategy to prevent PPM.

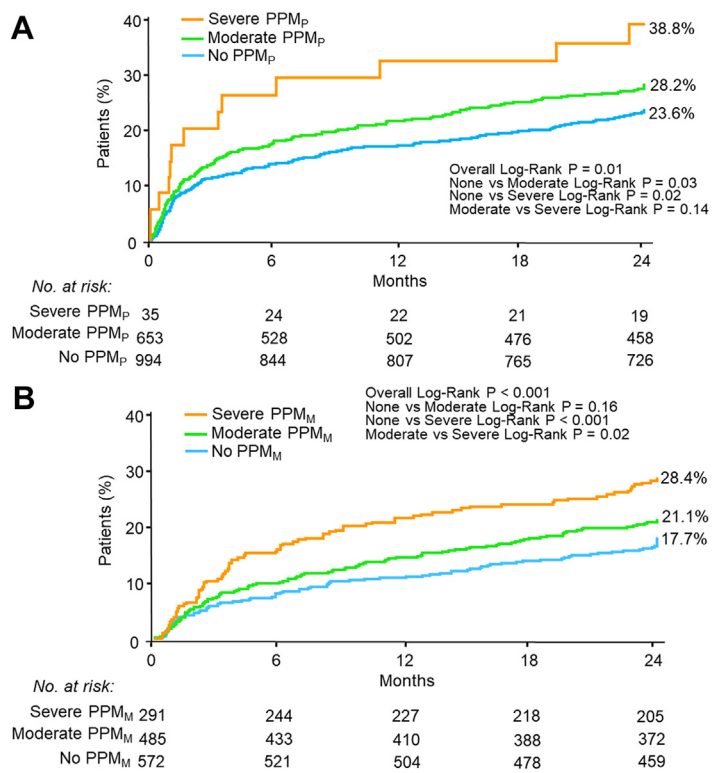

FIGURE 4 (A) All-cause death or heart failure rehospitalization, predicted patient-prosthesis mismatch (PPM_P). (B) All-cause death or heart failure rehospitalization, measured patient-prosthesis mismatch (PPM_M).

ROLE OF TAVR IN PREVENTING PPM. TAVR may provide a potential alternative to avoid PPM in those with severe aortic stenosis. Choosing TAVR in young patients to prevent PPM may be preferable when AE techniques may be risky to perform (eg, those with severe mitral annular calcification). However, use of TAVR is reliant on the performance of a preoperative computed tomography scan, which would notify the surgeon that an aortic root reconstruction technique will be required.

Comparisons between TAVR and surgery have generally concluded that less PPM occurs in TAVR in the short- and mid-term.[4,16-19] Because most initial TAVR studies reported PPM_M, rates of moderate to severe PPM were very high in SAVR compared with prior surgical series, which had traditionally used PPM_P. In PARTNER 1, Pibarot and associates[16] noted 28.1% severe PPM rate in SAVR patients compared with 19.7% in TAVR patients. Severe PPM was an independent predictor of

2-year mortality in the SAVR cohort (hazard ratio, 1.78, $P = .041$), but not in the TAVR cohort (hazard ratio, 0.58, $P = .11$). Most recently, Ternacle and coworkers[4] compared predicted and measured EOAi in patients undergoing SAVR or TAVR in the PARTNER 2 trial and noted that severe PPM for SAVR patients was 1.2% in the predicted EOAi method compared with 23.6% in the measured EOAi. The incidence of severe PPM using both the measured and predicted EOAi was lower in patients undergoing TAVR compared with SAVR ($P < .001$). While severe PPM was associated with increased all-cause mortality and re-hospitalization in SAVR patients, it was not so for TAVR.[4] Similarly, this relationship was also noted when comparing SAVR to self-expanding transcatheter AV in annular perimeter-derived diameters: small (<23 mm), medium (23 to <26 mm) and large (≥26 mm).[18] They noted that TAVR had significantly less PPM than SAVR in small and medium annuli ($P < .001$), with no difference in the incidence of PPM in large annuli ($P = .1$).[18] Some of this difference may be explained by use of the predicted or measured PPM, as well as the low incidence of severe PPM in small studies, short length of follow up, and underpowering for outcome events. We strongly advocate for a heart team assessment for the choice of TAVR or SAVR that remains critical not only for the initial procedure, but for future interventions.

STUDY LIMITATIONS. The percentage of missing data was close to 25% with the measured EOAi method but <10% with the predicted EOAi method. The incidence of PPM in the PARTNER trials may not reflect real-life clinical practice, as it was conducted in selected centers and included highly selected patients. Hence, the incidence of both severe measured and predicted PPM might be higher in real-life practice vs in these trials.

CONCLUSIONS. In this large study with adjudicated core laboratory assessment, our findings suggest that EOAi measured by TTE overestimates the incidence of true-severe PPM after SAVR when compared with predicted EOAi measurements. Although rare, severe PPM is associated with increased mortality and rehospitalization at 2 years. Intraoperative techniques to prevent PPM are imperative not only for the index procedure, but also for future transcatheter valve-in-valve options.

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DISCLOSURES

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